

Cooperated Truck-Drone Routing With Drone Energy Consumption and Time Windows

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Abstract—Considering customer time windows, multiple trucks, each equipped with a multi-visit drone, are employed. For realism, the drone energy consumption and the impact of payload variation on the energy consumption rate are considered. A Mixed Integer Linear Programming (MILP) model is developed to formulate the problem. A novel concept called “Segment” is introduced to promote the cooperation between trucks and drones. Based on this, a heuristic is designed where the drone and truck routes are constructed synchronously. The variable neighborhood search algorithm is integrated with simulated annealing to enhance solutions further. The effectiveness of the proposed algorithm is validated through both Solomon instances and practical cases.

Index Terms—Truck and drone, vehicle routing, delivery, time windows, heuristic, variable neighborhood search.

I. INTRODUCTION

IN RECENT years, cooperation delivery with trucks and drones has been proven to significantly improve efficiency while also providing cost savings [1], [2], [3], [4]. With longer driving range and higher load capacity, trucks can serve as mobile depots, where drones can replace the battery and reload cargo. Moreover, drones are less restricted by ground traffic and can perform better in some areas that are challenging for trucks to access, such as some places with heavy congestion or inconvenient transportation. This cooperation enables simultaneous task execution, significantly reducing overall delivery time.

Several industrial companies have embraced the applications of truck-drone delivery systems, such as “HorseFly” applied by UPS [5], which further validated the efficiency of this cooperation. Meanwhile, researchers have shown great interest in operational challenges, especially in optimizing delivery

routes. Since Murray and Chu first proposed the Flying Sidekick Traveling Salesman Problem (FSTSP) [6], this problem has been expanded with more complex constraints, such as a single truck equipped with multiple drones [7], [8], or further considering the drone pickup [9], [10], [11].

Most existing studies on truck-drone delivery systems assume that drones can only serve one customer per flight. However, drones’ flight endurance and payload capacity have been significantly improved. For example, the DJI Fly-Cart 30 Drone boasts a maximum payload of 30kg [12], while the majority of Amazon packages weigh under 5 pounds [13], making it feasible for drones to carry multiple parcels per trip. There have been multiple studies investigating the applications of multi-visit drones [8], [14]. However, many multi-visit studies overlook the impact of payload weight on energy consumption, which may lead to costly or infeasible routes in applications [15], [16]. Thus, it is essential to consider the variation in energy consumption rate caused by payload changes during drone delivery.

Another factor that has not been fully considered is the time preference of customers. Generally, customers expect parcels to be delivered when they are available, and express companies provide flexible delivery time options to boost customer satisfaction. Thus, it is crucial to integrate customer time windows into delivery route optimization.

Motivated both by practical applications and the theoretical gap existing in the current literature, a novel Vehicle Routing Problem with Time Windows and Multi-visit Drone (VRPTW-MD) is studied, where customer time windows and the effect of drone payloads on energy consumption are both considered. Fig. 1 illustrates an example of VRPTW-MD, with specific details of the yellow route presented in Fig. 1(b). It can be observed that the energy consumption rate of the drone gradually decreases as the number of parcels is reduced, which has also been supported by existing study [8].

The VRPTW-MD problem introduces a higher level of realism but also brings much complexity. The cooperation between trucks and drones, under the dual constraints of customer time windows and drone endurance, presents particular challenges. On one hand, the truck must strike a balance on the time to launch the drone. Releasing the drone too early results in unnecessary energy consumption during waiting, while releasing it too late risks missing subsequent customers. On the other hand, when retrieving the drone, the truck must ensure timely recovery within the drone endurance and without violating the rendezvous node’s time window. It becomes evident that

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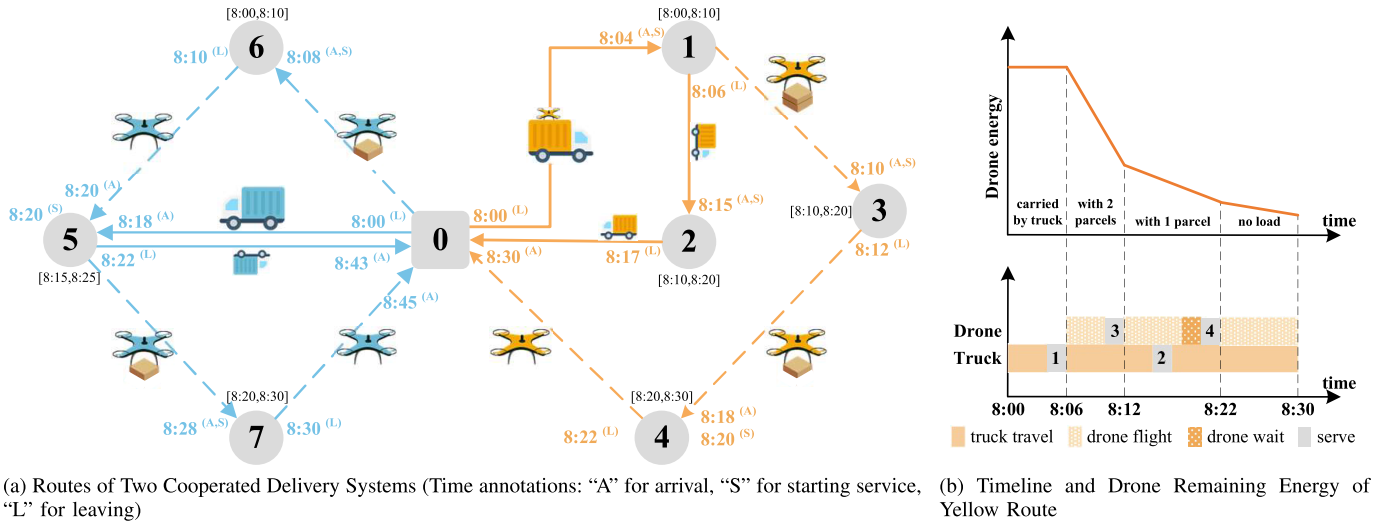


Fig. 1. Example of VRPTW-MD.

the VRPTW-MD favors the simultaneous operation of both vehicles.

In light of this, we introduce the concept of "Segment" and develop a Heuristic Based on Segment (HBS). This heuristic aims to achieve synchronized operation of the two vehicles, and addresses the limitations of previous two-stage heuristics in tackling this problem. A Variable Neighborhood Search algorithm combined with Simulated Annealing (VNS-SA) is further employed to enhance the solution quality. Both Solomon instances and practical cases are used to prove the proposed algorithms.

In summary, the contributions of the paper are as follows:

- 1) We investigate a novel truck-drone routing problem, where the impact of payload changes on energy consumption rates and customer time preferences are both considered.
- 2) We introduce an innovative concept of "Segment" is proposed, and develop a heuristic based on Segment enabling the simultaneous construction of drone routes and truck routes.
- 3) We merge the variable neighborhood search algorithm with simulated annealing, and design new operators based on the problem characteristics.
- 4) We propose a comprehensive benchmark for VRPTW-MD, including cases of varying scales and detailed parameters for trucks and drones.

The paper is organized as follows. Section II summarizes related works. Problem formulation and modeling are provided in Section III. Section IV presents the proposed algorithms and Section V shows the experimental results. Finally, conclusions are drawn in Section VI.

II. LITERATURE REVIEW

There has been considerable literature on the cooperated routing of trucks and drones. Multiple applications are explored, such as delivery, surveillance [17], [18], search [19], and area coverage [20]. Several surveys have explored this

cooperation from various perspectives, such as drone applications [3], [4] or two-echelon scheme [21]. This section focuses on two specific aspects: the Vehicle Routing Problem with Drones (VRP-D), and the cooperated truck-drone problem considering time-related factors.

There are two fundamental problems with the routing problem of cooperated truck and drone: Flying Sidekick Travelling Salesman Problem (FSTSP) proposed by Murray and Chu [6], and Travelling Salesman Problem with Drone (TSP-D) proposed by Agatz et al. [22]. In these two problems, only one truck and one drone are involved, and the drone can serve only one customer during a flight. These problems have been formulated as Integer Programming [23] or Mixed-Integer Linear Programming [24], [25]. A variety of exact algorithms [26], [27] and heuristic algorithms [28], [29], [30] have also been applied to solve such problems.

When multiple trucks, each equipped with a drone, perform tasks simultaneously, the planning problem is commonly referred to as a Vehicle Routing Problem with Drone (VRP-D). Poikonen et al. [31] first proposed the VRP-D and conducted analysis on two parameters: the number of drones per truck and the speed of the drones relative to that of the truck. Schermer et al. [32] built a novel mixed-integer linear programming model to obtain exact solutions. Branch-and-cut algorithm [33] and adaptive large neighborhood search (ALNS) algorithm [34] have also been applied to solve this problem.

There are various variations and extensions of the VRP-D. One of the extensions is to allow each truck to be equipped with multiple drones, which has been solved by the constraint-based hybrid pointer network model [35] and improved variable neighborhood search (VNS) algorithm [36]. Besides, Wang and Sheu [37] extended this problem by allowing drones to land on other trucks. Another extension is considering drone energy consumption. Multiple studies [15], [16] have proven that the drone payload would significantly affects its energy consumption rate. Most of the literature [8], [28] assumes that the energy consumption rate

is positively correlated with the payload, while a few studies [38] consider more detailed nonlinear formulas. Besides, Das et al. [39] explored the multi-objective optimization problem and developed a Collaborative Pareto Ant Colony Algorithm. Some papers [40], [41], [42] also explored the possibility of drones working with public transport vehicles or autonomous vehicles.

The time window has always been a crucial factor in the field of route planning [43], [44]. Consequently, several time-related factors have been considered, including not only customer time windows but also varying truck speeds during different time periods [45], [46]. However, introducing the time windows further fragments the solution space and significantly increases the solving complexity. Some previous studies have simplified or restricted the behaviors of vehicles. For instance, trucks are required to wait in place after launching the drone [47], [48], or recover the drone immediately at the next node from the launch node [35]. Additionally, the consideration of soft time windows [39], [49] relaxes the time constraints to some extent. This adjustment does not alter the problem's solution space but modifies the objective function, thereby shifting the optimal solution within the solution space.

When focusing on hard time windows, most studies [50], [51], [52] assume that the drone can only serve one customer in one flight. Although some papers [14] discuss scenarios where the drone can carry multiple parcels, they typically employ the maximum flight time as a measure of drone endurance, disregarding the effect of payload changes on energy consumption. Furthermore, some studies [53] allow the drone to land and wait without considering the energy consumed during the waiting period, thereby relaxing the constraint on the truck's arrival time at recovery nodes. In practical applications, these simplifications may lead to an overestimation of drone endurance, rendering the solutions impractical.

Considering the challenges posed by customer time windows in problem-solving, it may not be realistic to adopt exact algorithms to solve the problem. Most exact algorithms need to solve more than half an hour when dealing with more than 20 customers [14], [52]. As a result, many research on VRP-D with time windows relies on heuristics to generate feasible solutions quickly [47], [50]. Various heuristic strategies are designed based on problem characteristics. Furthermore, intelligent metaheuristic, such as genetic algorithms [49], [53], VNS [50], and ALNS [47], are applied to enhance solution quality. However, due to the highly discrete nature of the solution space, there remains an urgent need to expedite the search process and efficiently discover higher-quality solutions.

From the above review, we can see that there still exists a gap in the route planning problem for cooperated trucks and drones, especially when considering time windows, owing to the inherent complexity of the problem. To bridge this gap, this study first defines the specific details of cooperated delivery. The energy consumption of drones and the impact of drone payload on the energy consumption rate are also considered. Furthermore, in order to enhance the speed and quality of solutions for large-scale problems, a concept of "Segment" is

introduced, and corresponding heuristic and search algorithm are designed.

III. PROBLEM FORMULATION

This section provides a formal description of the Vehicle Routing Problem with Time Windows and Multi-visit Drone (VRPTW-MD) in Section III-A. Decision variables and the objective are described in Section III-B, and the detailed MILP formulation is available in supplementary materials [54].

A. Notations and Problem Description

The VRPTW-MD can be formally described over the directed graph (V, E) , with V as the node set and E as the arc set. Formally, $V = N \cup \{0\} \cup \{n+1\}$, where 0 and $n+1$ represent the depot, $N = \{1, \dots, n\}$ represent the customer nodes. E is defined as $\{(i, j) | i, j \in V, i \neq j\}$.

A fleet of cooperated delivery systems $K = \{1, \dots, k\}$, each consisting of a truck and a mounted drone, is provided to serve customers. The truck is equipped with sufficient drone batteries and corresponding charging devices, ensuring that drones can take off with a fully charged battery. A fixed cost (λ_1) is incurred when a cooperated delivery system is activated. The truck and drone have maximum weight capacities denoted as Q^G and Q^U respectively. Additionally, the drone has a self-weight of w^U and a maximum battery capacity of θ . Since the truck can only travel on the ground road while the drone can fly straight in the air, the traveling time of two vehicles along the arc (i, j) is different, denoted as t_{ij}^G and t_{ij}^U , respectively.

Both trucks and drones can serve customers, while each customer $i \in N$ can only be served exactly once. A package of weight w_i needs to be delivered, and the delivery operation will take a service time s_i^U or s_i^G if the customer is visited by a drone or a truck, respectively. A time window $[e_i; l_i]$ is associated with customer $i \in N$, where e_i and l_i represent the earliest and the latest time when the customer can be served.

As for the vehicle behaviors at customer nodes, a specific ordering policy is implemented: (1) the truck/drone arrives; (2) the truck retrieves the drone if the customer is a retrieval node; (3) the truck/drone serves the customer; (4) the truck/drone leaves as soon as finishing service; (5) the truck launches the drone as it leaves if the customer is a launch node. If the drone reaches the retrieval node before the truck, it is allowed to wait and hover until the truck arrives. This waiting period also consumes energy, and the truck must retrieve the drone before it runs out of energy.

Several assumptions are considered to specify the operation process of the VRPTW-MD:

- (1) It is assumed that the cooperated delivery systems are homogeneous.
- (2) The truck is typically equipped with enough fuel or battery capacity to manage daily delivery tasks, which allows us to assume that the truck has an unlimited travel range while being limited by its maximum capacity. The drone is constrained by both endurance and payload capacity.
- (3) It is assumed that both the truck and the drone travel at a constant speed specific to each of them.

TABLE I
DECISION VARIABLES

Name	Description
$z_{ik}^C \in \{0, 1\}$	Indicates whether customer i is visited by both the k -th truck and the k -th drone
$z_{ik}^G \in \{0, 1\}$	Indicates whether customer i is only visited and served by the k -th truck
$z_{ik}^U \in \{0, 1\}$	Indicates whether customer i is visited by the k -th drone
$y_{ijk} \in \{0, 1\}$	Indicates whether the k -th truck travels along arc $(i, j) \in E$
$t_i^{G,A} \geq 0$	The truck's arrival time at node i
$t_i^L \geq 0$	The truck and drone's departure time from customer i
$t_k^L \geq 0$	The k -th truck and drone's departure time at the depot
$z_{ik}^U \in \{0, 1\}$	Indicates whether customer i is served by the k -th drone
$x_{ijk} \in \{0, 1\}$	Indicates whether the k -th drone travels along arc $(i, j) \in E$
$t_i^{U,A} \geq 0$	The drone's arrival time at node i
$t_i^R \geq 0$	The retrieval time at customer i
$t_k^R \geq 0$	The retrieval time of the k -th drone at the depot
$t_i^S \geq 0$	The service start time at customer i
$0 \leq p_i^U \leq \theta$	The remaining battery of the drone at point of departure from node i
$w_i^U \geq 0$	The total package payload when a drone departs from node i
$w_k^U \geq 0$	The total package payload when the k -th drone departs from the depot
$h_i^L \in \{0, 1\}$	Indicates whether customer i is the launch node
$h_i^R \in \{0, 1\}$	Indicates whether customer i is the retrieval node

(4) Each drone can only return to the truck from which it was launched.

(5) The truck must leave the customer upon launching the drone, that is, the truck cannot wait at the same node.

(6) To ensure the safe retrieval of the drone, it is required that the truck can only start service after recovering the drone.

B. Decision Variables and Formulation

A series of decision variables are defined to solve the problem, of which a summary is provided in Table I.

Three binary decision variables, z_{ik}^C , z_{ik}^G , and y_{ijk} , define whether the customer $i \in N$ is visited by only truck, only drone, or both two vehicles.

Multiple continuous decision variables are defined to describe the schedule of trucks and drones, such as t_k^L , t_k^R , $t_i^{G,A}$, and $t_i^{U,A}$. y_{ijk} and x_{ijk} describe whether the truck or the drone travels along arc (i, j) , respectively. It is noted that if the truck carries the drone along arc (i, j) , x_{ijk} will also be set to 1, which ensures a complete drone route both from and to the depot. Continuous variables p_i^U and w_i^U represent the remaining energy and payload of the drone when leaving node i . For launch nodes, p_i^U will be directly set to θ . h_i^L and h_i^R indicate if node i is a launch node or a retrieval node, respectively.

After defining decision variables, the objective of VRPTW-MD can be described. The VRPTW-MD involves planning a set of routes to satisfy a set of customers while minimizing the total cost. The total cost consists of three parts: a) to minimize the time required for all delivery systems (each consisting of one truck and one drone) to deliver assigned parcels and return to the depot; b) to minimize the longest delivery time among all delivery systems to ensure balanced

workloads for the drivers, c) to minimize the number of delivery systems. The objective can be formulated as:

$$\min \lambda_1 \times \sum_{k \in K} (t_k^R - t_k^L) + \lambda_2 \times \max_{k \in K} \{(t_k^R - t_k^L)\} + \lambda_3 \times \sum_{k \in K} \sum_{j \in N} y_{0jk}, \quad (1)$$

where λ_1 measures the proportion of total delivery time, λ_2 represents the importance of workload balance among different trucks, and λ_3 represents the usage cost. The specific value will be described in the experiments.

The remaining part of the MILP formulation is accessible in supplementary materials [54].

IV. ALGORITHM

Considering multiple complex constraints in the routing problem with trucks and drones, directly solving the MILP model may not be able to handle large-scale cases. Hence, there is a need to design a heuristic algorithm that can quickly provide solutions. However, incorporating time windows poses a significant challenge to the heuristic approach. The existing two-stage heuristic, which constructs truck routes first and then modifies them into cooperated routes, struggles to collaborate effectively between trucks and drones.

In light of the problem characteristics of VRPTW-MD, we propose a constructive heuristic based on the concept of Segment, then a variable neighborhood search algorithm integrated with a simulated annealing algorithm is designed to enhance the solution quality.

A. Definition of Segment

In the VRPTW-MD, either the truck or the drone is allowed not to start service immediately after arriving at a customer, while it must leave promptly upon completing the service. Thus, when defining the Segment, the drone launch/retrieval node is selected as the dividing node. The latest arrival time of two vehicles at the node is the end time of the previous Segment, whereas the leave time is the start time of the next Segment. Specifically, the Segment can be divided into two types based on whether it includes a drone route:

(1) Cooperated Segment: comprising a truck route segment and a complete drone route, which starts from the drone launch node and ends at the drone retrieval node. Fig. 2 contains two cooperated Segments, Segment 1 and Segment 2.

(2) Pure Truck Segment: only including a truck route segment, which starts from the retrieval node of the last cooperated Segment (or depot) and ends at the launch node of the next cooperated Segment (or depot). Segment 3 in Fig. 2 is a Pure truck Segment.

Considering the drone route with its corresponding truck route segment as a Segment, the complex cooperated route can be transformed into a linear connection of multiple Segments.

B. Heuristic Based on Segment

In the Heuristic Based on Segment (HBS), a new route is generated as long as there remain unvisited customers.

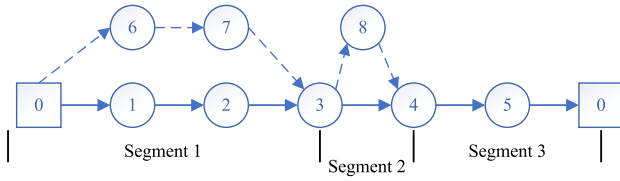


Fig. 2. Illustration of segments.

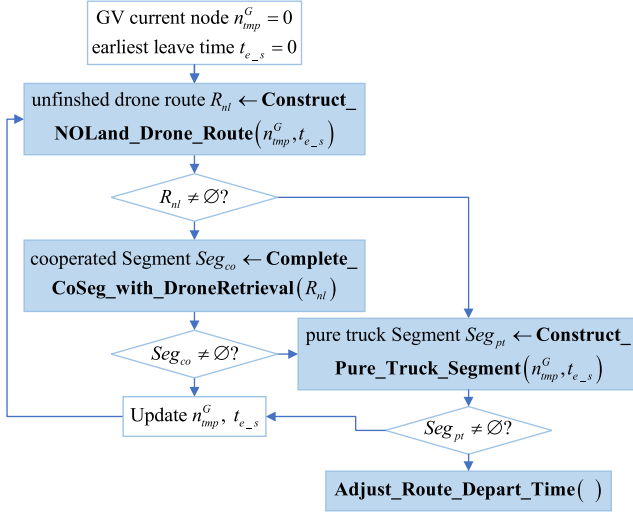


Fig. 3. Flowchart of generating a cooperated route.

The algorithm flowchart for generating a cooperated route is illustrated in Fig. 3.

When planning the route for a truck and its drone, efforts are made to construct cooperated Segments as much as possible, where both vehicles serve customers synchronously. This is achieved through two modules: **Construct_NOland_Drone_Route**, which builds an unfinished drone route without considering landing, and **Complete_CoSeg_with_DroneRetrieval**, which plans the corresponding truck route segment while considering the drone endurance and customer time windows. If cooperation between the two vehicles fails, the module **Construct_Pure_Truck_Segment** is attempted, making the truck transport the drone to the next customer node. Whenever the route state changes, a new attempt is made to build the cooperated Segment. When the truck must return to the depot, the departure time of this route is optimized using **Adjust_Route_Depart_Time**.

1) *Construct Drone Route Without Landing*: Multiple constraints must be considered in a cooperated Segment, including drone payload, endurance, and customer time windows. Building two routes within the same Segment simultaneously can be challenging. Moreover, given the clear advantages of drones over trucks in speed and cost, it is generally beneficial to have drones visit more customers. Thus, the heuristic aims to build an unfinished drone route initially, making the drone serve as many customers as possible without considering landing. The pseudo-code for the part is shown in Algorithm 1.

The drone launches from the current truck node, with the launch time initially set as the provided earliest start time t_{e-s} . The maximum delay is first calculated based on the time

windows of the launch node. When determining the next drone node, the remaining unvisited customers are first sorted based on the distance from n_{imp}^U . Each customer would be verified to check whether the drone can serve it within the designated time window, has the capacity to carry its parcel, and possesses enough battery power to reach it. Once the drone finds enough accessible customers, it stops calculating for other customers located farther away. Among the identified customers, the node at which the drone can complete the delivery earliest is selected as the next drone node. The maximum feasible delay and the best launch time are updated according to the time window of this customer.

2) *Complete Cooperated Segment With Drone Retrieval*: After obtaining an unfinished drone route through Algorithm 1, it is important to find a suitable customer node for drone retrieval with the corresponding truck route segment. The truck must recover the drone within the drone endurance and before the latest service time of the land node, which puts higher requirements on the cooperation of two vehicles. The pseudo-code is shown in Algorithm 2.

Given an unfinished drone route R_{nl}^U , the customer node and departure time for the truck are determined. Then, calculate the set of unvisited customers that the truck can serve after leaving node n_0 at the designated leaving time. These customers would be sorted according to their latest service time. Next, the subset that can be used for drone retrieval is determined by verifying whether the drone's battery can support it to reach the node and wait for retrieval. It also needs to ensure that the truck can still serve customers within the feasible time window after retrieving the drone. For each landing node, a truck route segment is built by inserting customers from set N_{imp}^G . This insertion should not violate any customer's time window constraints and not cause the drone to wait excessively. Among these cooperated Segments, the most balanced one would be

Algorithm 1 Construct No-Land Drone Route

Require: n_{imp}^G, t_{e-s}

- 1: $n_{imp}^U = n_{imp}^G, t_{launch} = t_{e-s}$
- 2: $delay_{max} = \left(l_{n_{imp}^G} + s_{n_{imp}^G} \right) - t_{launch}$
- 3: **while** exist accessible drone customers **do**
- 4: $count = 0$
- 5: Sort rest customers by distance to n_{imp}^G
- 6: **while** $count < MAX_Count$ **do**
- 7: $n_{imp} \leftarrow i_{th_nearest}$ unserved customer
- 8: **if** the drone can serve n_{imp} **then**
- 9: $n_{ear} \leftarrow$ if the drone leaves n_{imp} at the earliest
- 10: $count++$
- 11: **end if**
- 12: **end while**
- 13: **if** find n_{ear} **then**
- 14: Add n_{ear} into the drone route
- 15: $n_{imp}^U = n_{ear}$
- 16: Update $t_{launch}, delay_{max}$
- 17: **end if**
- 18: **end while**

Algorithm 2 Complete Cooperated Segment

Require: unfinished drone route R_{nl}^U

```

1: while true do
2:   launch node  $n_0 \leftarrow$  first node in  $R_{nl}^U$ 
3:    $N_{un}^G \leftarrow$  set of customers that truck can serve
4:    $N_{land}^U \leftarrow$  set of customers able to recover the drone
5:   if  $N_{land}^U \neq \emptyset$  then
6:     Sort nodes in  $N_{un}^G$  based on the latest serve time
7:     for  $n_q^U \in N_{land}^U$  do
8:        $R_q^G \leftarrow$  insert nodes in  $N_{un}^G$  between  $n_0$  and  $n_q^U$ 
9:        $Seg_q \leftarrow$  combine  $R_q^G$ ,  $R_{nl}^U$  and  $n_q^U$ 
10:    end for
11:    Select the most balanced cooperated Segment
12:    break
13:  else
14:    Delete last customer node in  $R_{nl}^U$ 
15:    if no customers served by the drone in  $R_{nl}^U$  then
16:      break
17:    end if
18:  end if
19: end while

```

selected, considering the arrival times of two vehicles and the time window of the land customer.

However, if no feasible landing nodes can be found for the current unfinished drone route, the last customer in the route R_{nl}^U will be deleted. The above steps are repeated until a cooperated Segment is generated or no more customers are served in R_{nl}^U .

3) *Construct Pure Truck Segment*: Due to the limited endurance of the drone, there may be situations where the drone cannot reach some distant customers. In such cases, the truck is required to carry the drone to some near nodes.

When constructing a truck Segment, feasible customers that the truck can serve are identified, considering constraints such as truck capacity and customer time window. Among these feasible customers, the one that the truck leaves earliest is selected. It aims to minimize the duration during which the truck works independently, and create more opportunities for collaboration. If the truck cannot access any customers, it will return to the depot.

4) *Adjust Route Depart Time*: According to the algorithms above, the departure time of cooperated Segment is depended on the drone route, which may not be optimal start time for the Segment. As shown in Fig. 4, the black and orange lines indicate that the truck and the drone leave node N_{depart} at times s_I and s_{II} , respectively. Although the black drone route does not encounter any customer waiting time, the truck must wait for customer N_3^G . In this case, delaying the leaving time from s_I to s_{II} can reduce the delivery time.

Furthermore, the best start time of the first Segment may not be the best start time for the entire route as well. Therefore, after obtaining a cooperated route, it is necessary to adjust the departure time from the depot.

When determining the optimal start time for a given route, it is necessary to calculate both the best start time and the

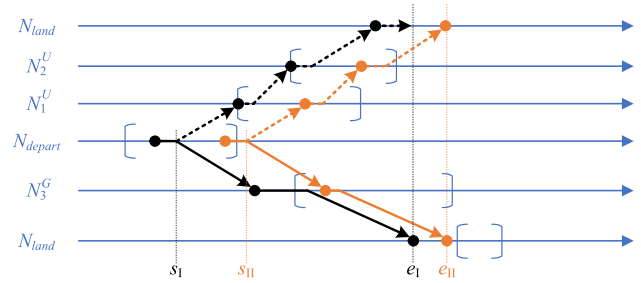


Fig. 4. Best start time of a cooperated segment.

latest start time for each Segment. Subsequently, the best start time for the entire route can be calculated by considering the route as a linear sequence of multiple Segments.

C. Variable Neighborhood Search With Simulated Annealing

Although the initial solution given by HBS can facilitate the cooperated delivery between trucks and drones, there is still room for improvement. A variable neighborhood search algorithm integrated with simulated annealing is proposed. We have designed three operators within this framework based on the problem characteristics. During the neighborhood operations, each potential new solution undergoes a feasibility check to ensure its practicality and effectiveness. We can rapidly assess the validity of new solutions by pre-computing feasible time windows for each Segment without the need for extensive re-computation.

1) *Operator 1: Split Segments*: When constructing the initial solution by HBS, a new combination of a truck and its drone is introduced only when the last route can no longer visit any additional customers. Therefore, the routes established at the beginning tend to have a longer duration in the initial solution, which is inconsistent with the expectation of balancing the working time of multiple trucks.

Initially, it is randomly selected whether to split the route with the longest duration or a random route. As illustrated in Fig. 5, Seg_1 is a pure truck Segment and Seg_2 is a cooperated Segment. When node N_m is contained in the latter Segment, three cases are displayed. Fig. 5(b) shows the simplest way where a pure truck Segment is added from the depot to node N_m . In the other two cases, the drone launch node is changed to the depot, and node N_m is inserted into either the truck route or the drone route. If the middle node N_m is contained in the former Segment, the new start node for Seg_2 can only be found between the depot and the first customer N_1 . Among all the above situations, two new routes with the shortest total time are selected.

Similarly, if Seg_1 is a cooperated Segment and Seg_2 is a pure truck Segment, the landing node of the drone route in Seg_2 needs to be determined during splitting. However, if both Segments cooperate, the computation becomes more complicated, and there may be cases where generating two feasible routes is impossible.

2) *Operator 2: Swap*: The swap operator can be classified into four types based on whether the same type of vehicle visits the selected two nodes and whether the two nodes are in the same Segment.

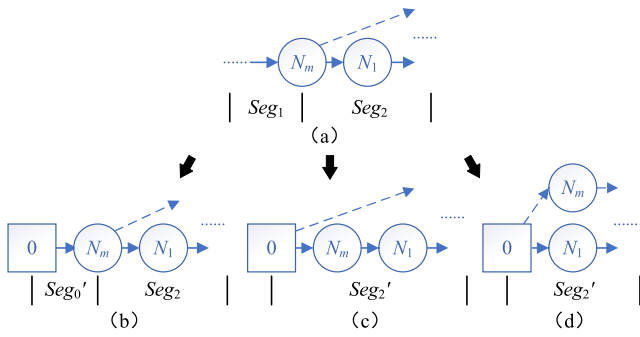


Fig. 5. Illustration of splitting segments.

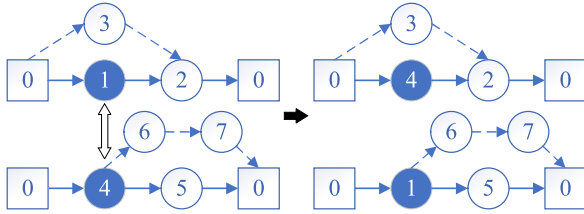


Fig. 6. Illustration of swapping two nodes visited by two trucks.

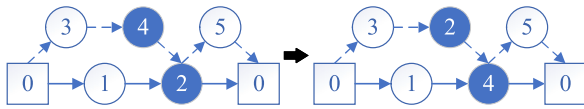


Fig. 7. Illustration of swapping heterogeneous nodes in the same segment.

(a) Since there must exist a sequence between the arrival times of two customer nodes in the same route (either drone or truck), it is challenging for two homogeneous nodes in the same Segment to satisfy their respective time window constraints after exchange. To reduce unnecessary calculations, this type of swap is not considered.

(b) When swapping two homogeneous nodes in different segments, only two customer nodes from two routes with overlapping time windows are considered. If the truck customers are selected, some modifications of drone routes may be involved. As shown in Fig. 6, when swapping customer 1 and 4, it is necessary to evaluate the feasibility of the drone route $\{1, 6, 7, 0\}$. It also should be examined whether the swapping will impact subsequent visits.

(c) It is essential to swap heterogeneous customers within the same Segment. The HBS algorithm can effectively cluster customers with similar time windows and close distances in one Segment. This type of exchange can potentially reduce the time consumption within the Segment while not causing a significant impact on the subsequent route. However, if the selected truck customer is also located in another Segment, such as customer 2 in Fig. 7, the second Segment should be changed and its feasibility must also be verified.

(d) Exchanging heterogeneous customers in different Segments is an effective way to expand the search range and enhance the diversity of solutions.

3) *Operator 3: Remove-Insert*: To increase the diversity of solutions in the search process, this operator employs the strategy of random removal. After randomly selecting a

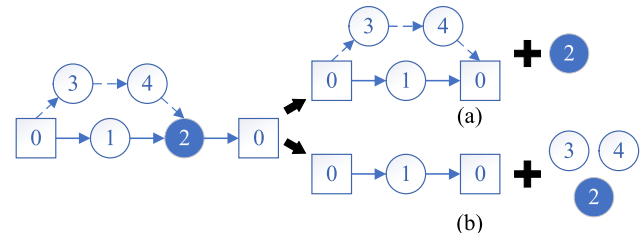


Fig. 8. Two cases of removing a truck node.

Segment, a node in this Segment would be chosen. Specifically, if the Segment is a cooperated One, the waiting situation at the recovery node is evaluated, and the node to be deleted is then selected from the longer route.

After determining the removed node, further judgment is conducted. If the customer is visited by a truck and used for drone launch/recovery, the operator will first try to find an alternative node for the drone route, such as Fig. 8(a). If the attempt fails, the entire drone route will be deleted, and the original cooperated Segment will be modified to a pure truck Segment, as shown in Fig. 8(b).

When inserting a removed customer, three insertion situations are considered: inserting the customer into an existing truck route or drone route, adding a new drone route within a pure truck Segment, or scheduling an additional truck to visit the customer. The one that takes the least time would be adopted. If multiple customers have been deleted, the nodes are randomly sorted and then inserted one after the other.

V. EXPERIMENTS

In this section, we first solve Solomon cases using the MILP model and the proposed HBS-VNS-SA, and then benchmark the HBS-VNS-SA against the exact algorithm by Yin et al. [14]. Additionally, three scales of actual instances are presented and applied to further explore the algorithm's performance. Lastly, sensitivity analysis is conducted on some critical factors, including the average length of time windows and the service time, as well as some parameters related to the drone performance. A detailed description of the experimental equipment, as well as the instances used in Section V-A.1 and V-B, are available online in supplementary materials [54].

A. Experiment With Exact Algorithms

1) *Experiment on Solomon Instances*: As the VRPTW-MD is a newly introduced problem, a new set of test instances is derived from the widely used Solomon instances [55]. The first 6 or 8 customers from each Solomon case are selected, resulting in 112 small-scale instances. The truck and drone parameters are designed based on existing parameters, as indicated in the third column of Table II. Regarding the three parameters in the objective function, the values of parameters λ_1 , λ_2 , and λ_3 are temporarily set as 1, which will be adjusted according to actual applications when it comes to actual cases.

Both the CPLEX solver and the proposed HBS-VNS-SA algorithm are applied to small instances. Detailed results per instance are posted online in supplementary materials [54]. Table III summarizes the comparison results. Results indicate

TABLE II
PARAMETERS OF TRUCK AND DRONE

Vehicle	Characteristic	Solomon Cases	Actual Cases
Truck	speed v^G	1	40km/h
	payload capacity Q^G	1000	500kg
drone	speed v^U	2	50km/h
	self-weight w^U	30	5kg
	payload capacity Q^U	50	5kg
	battery capacity θ	2000	5000mAh
	coefficient α	1	2.5mA/g

that the CPLEX solver solved all test instances with six customers optimally. However, it could only find the optimal solution for 20 instances out of 56 cases with eight customers. The computational time required to solve the instances increased significantly when the number of customers increased from six to eight. Regarding the proposed HBS-VNS-SA algorithm, it found the optimal solutions for all instances with six customers. For most cases with eight customers, the algorithm provided more optimal solutions compared to those obtained by CPLEX. In each of the ten runs for 20 instances of C1 and R2, the HBS-VNS-SA algorithm consistently achieved the same BSF solution. The average gap was less than 1% for most instances, whether there were six or eight customers. This indicates that HBS-VNS-SA can perform consistently well in solving small-scale instances. Moreover, the computation time of the proposed algorithm was significantly shorter than that of CPLEX. The algorithm solved the instances with six and eight customers within 2 seconds and 3.5 seconds, respectively.

2) *Experiment on Similar Problem:* Yin et al. [14] have introduced a closely related problem. The main difference is that they only consider the maximum flight time of drones, while overlooking factors such as energy consumption during waiting periods and the influence of drone payload on energy consumption. The total delivery cost, including the use cost and travel cost of all trucks and drones, is minimized. To deal with this, they designed an efficient branch-and-price-and-cut (BPC) algorithm which could find the optimal solution for most of cases within 35 customers.

To validate the efficacy and versatility of the proposed HBS-VNS-SA, several modifications were made to solve several cases involving 35 customers from the literature [14]. The experimental results are shown in Table IV.

For instances with 35 customers, the average computation time for HBS-VNS-SA is around 13 to 17 seconds, which is significantly lower than that of BPC. Moreover, despite the slight increase in solutions by HBS-VNS-SA compared to BPC's optimal solutions, the average gap remains well within an acceptable range of 3%. This indicates that our algorithm can provide near-optimal solutions while drastically improving computational efficiency.

B. Experiment on Actual Instances

1) *Case Description:* Three scales of cases are built based on the road network and customer nodes in Changsha, China. The left part of Fig. 9 displays main roads within the second

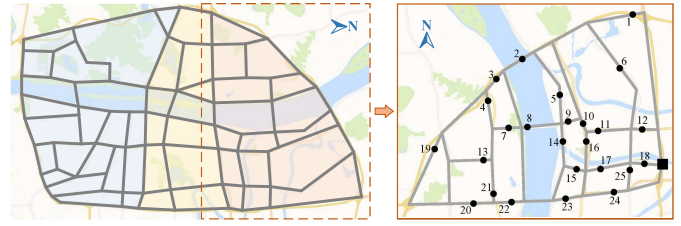


Fig. 9. Illustration of road network and a Small-scale instance (S-04).

ring of Changsha, which is roughly divided into three parts. Small-scale cases with 25 customers are generated in the red subarea (50km²). Medium-scale cases involve red and yellow subareas (85km²), containing 50 customers. Large-scale cases cover all the subareas (140km²), including 100 customers.

Specifically, an intersection node in the road network is first selected as the depot. Customers' locations are generated along the roadside, accompanied by randomly assigned parcel weights and time windows. According to the previous study [28], the package weight is set within 5kg, with 90% of the packages in the 0-3kg range. The working time for the depot spans from 8:00 a.m. to 8:00 p.m. The average duration of time windows is set to 1 hour. The service time for each customer is generated in the range of 15 minutes.

The parameters related to trucks and drones are defined according to typical ones in practical use. Specifically, two rotary-wing drones, Ehang Falcon B [56] and DJI Matrice 3 [57], are referred. The drone weighs about 5 kg with a maximum payload of 5 kg. With a battery capacity of 5000mAh and endurance of about 10 km under maximum payload, the coefficient α is calculated as 2.5mA/g. The truck is set to have a capacity of 500 kg and an average speed of 40 km/h. The parameters are also reported in Table II. Moreover, the flight distance of the drone between two nodes is the Euler distance, while the travel distance of the truck is calculated according to the connectivity of the road network.

The cost parameters in objective are referred to study [58]. The travel cost of the delivery system (λ_1), including the salary of deliveryman, truck traveling cost, and drone flight cost, is calculated to be approximately 29.3 €/per hour. The usage cost of a delivery system (λ_3) is estimated to be approximately 40 €. As for the value of λ_2 , it mainly depends on the importance placed by the delivery company on balancing the workload. To ensure that this aspect plays a significant role without being excessively skewed, λ_2 is set to 30.

2) *Result of Actual Cases:* Further experiments are conducted on 30 actual cases in different scales. The HBS-VNS-SA algorithm was executed ten times on each case, and the best solution achieved is shown in Table V.

The HBS-VNS-SA algorithm can effectively solve cases in different scales. For large-scale cases involving 100 customers, the HBS-VNS-SA can provide solutions within 40s. By analyzing the solutions generated by the HBS-VNS-SA, the overall objective cost of three scales of cases is about 432 €, 696 €, and 1199 €, respectively. In the small-scale instances, it achieved efficient delivery times and costs, averaging 4.59 hours for total delivery, 1.13 hours for maximum

TABLE III
SUMMARY RESULTS OF SMALL-SCALE INSTANCES

N_{cus}	Inst.	CPLEX			HBS-VNS-SA				
		Num_{opt}	Obj_{avg}^{CPLEX}	T_{avg}^{CPLEX} (s)	Obj_{avg}^{BSF}	Obj_{avg}^{AVG}	Gap_{avg}^{BSF}	Gap_{avg}^{AVG}	T_{avg}^{AVG} (s)
6	C1	9/9	877.22	35.02	877.22	877.22	0.00%	0.00%	0.73
	C2	8/8	978.41	44.18	978.41	979.09	0.00%	0.09%	0.71
	R1	12/12	168.41	184.96	168.41	168.45	0.00%	0.31%	1.30
	R2	11/11	169.00	317.04	169.00	169.00	0.00%	0.00%	1.33
	RC1	8/8	218.62	182.99	218.62	220.01	0.00%	0.72%	0.85
	RC2	8/8	245.10	288.77	245.11	246.67	0.00%	1.09%	1.12
8	C1	6/9	1095.16	1254.45	1095.16	1095.16	0.00%	0.00%	1.22
	C2	5/8	1226.45	1388.08	1223.31	1223.49	-0.26%	-0.24%	1.24
	R1	3/12	221.69	2831.28	204.28	205.89	-10.29%	-6.76%	1.90
	R2	2/11	241.248	3042.44	207.50	208.66	-16.52%	-14.31%	1.95
	RC1	2/8	257.31	3137.71	246.09	250.90	-5.76%	-2.45%	1.33
	RC2	2/8	291.33	2861.69	273.73	278.41	-8.58%	-5.01%	1.93

¹ N_{cus} : the number of customers; Inst.: six series of Solomon instances.

² Num_{opt} : the ratio of optimal solutions found; Obj_{avg} : the average objective value; “BSF”: best solution found by HBS-VBS-SA.

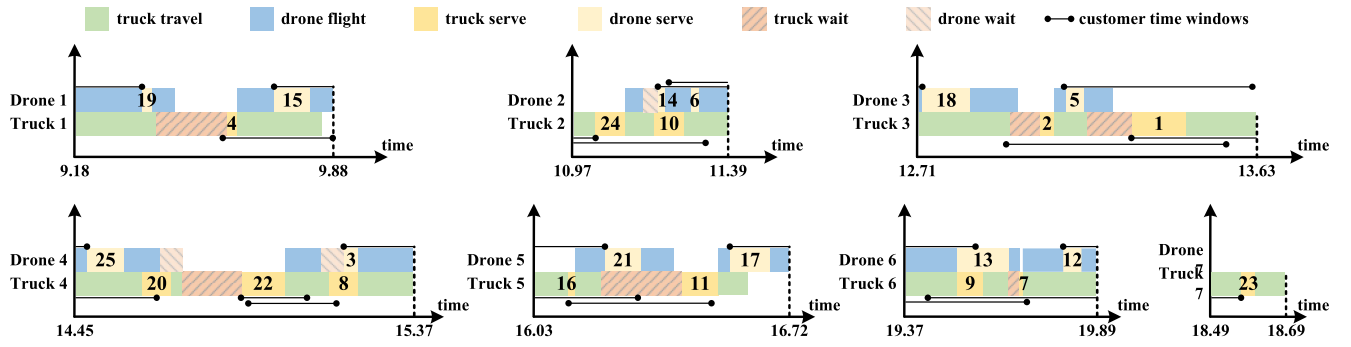


Fig. 10. Timeline of solution of a Small-scale case (S-04).

TABLE IV
COMPARISON RESULTS WITH BPC ON A SIMILAR PROBLEM

N_{cus}	BPC		HBS-VNS-SA		
	Cost	CPU(s)	$Cost_{avg}$	CPU_{avg} (s)	Gap
35	260.03	386	260.16	14.62	0.05%
	258.51	1136	265.36	13.95	2.65%
	258.22	1523	259.39	16.86	0.45%
	260.25	1077	261.01	13.14	0.29%
	258.31	1508	257.26	13.97	-0.41%
	256.56	2004	261.25	16.96	1.83%
	249.26	2628	250.65	17.01	0.56%
	258.25	1517	261.34	14.59	1.20%

delivery, and 432.40 € in objective cost, using 6.6 trucks on average. As the scale increased to medium and large, the algorithm adapted effectively, managing longer delivery times and higher costs while maintaining efficient resource allocation. Notably, as the number of customers rises from 25 to 50, there is a significant rise in the average number of trucks from 6.6 to 9.2, indicating a 39% increase. However, when the number of customers reaches 100, the impact on average working time becomes more pronounced. With the increase in customers from 50 to 100, the maximum working time doubles from 1.68 to 3.37. In scenarios with many customers, increasing the allocated working time for each truck may contribute more significantly to reducing the total cost.

Fig. 9 demonstrates a small-scale case (S-04) comprising one depot (represented by the square) and 25 customers (indicated by dots). Its best solution achieved by HBS-VNS-SA involves seven routes. Each route is plotted into a timeline, which is illustrated in Fig. 10. Among the 25 customers, drones visit 12 customers, while the remaining are served by trucks. Despite some routes appearing to have slightly extended waiting periods, such as Truck 1 and Truck 5, it's important to note that these brief pauses at customer locations can actually help reduce travel time to and from the depot. This, in turn, contributes to minimizing the overall delivery duration. Moreover, postponing departures from certain customers enables drones to efficiently service a greater number of locations. In addition, the perfect cooperation between trucks and drones is evident in multiple routes. For instance, Truck 6 experiences only a minimal wait time of 0.03 hours at customer 7, and for the majority of its journey, the truck and drone operate in perfect synchronicity. This showcases the efficiency and optimization capabilities of the HBS-VNS-SA algorithm in real-world logistics scenarios.

3) *Comparison Experiment*: As the feasible time of Segments is pre-calculated, the introduction of “Segment” can avoid consuming excessive time in the search process. To verify the effectiveness of “Segment”, an algorithm called HBS-VNS-SA-noSeg is proposed, which does not utilize existing Segment information during the solution verification.

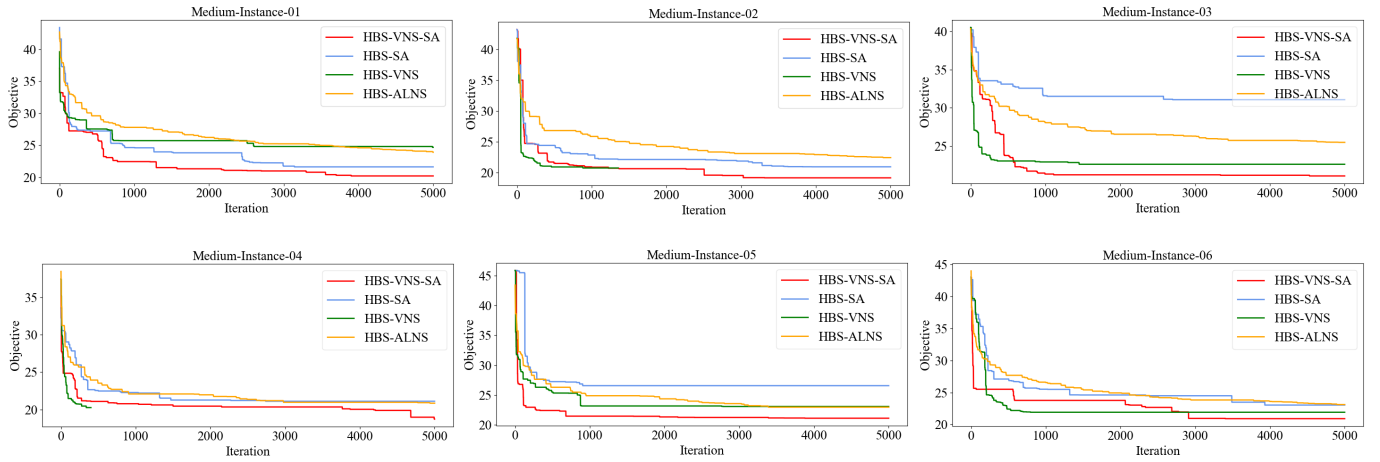


Fig. 11. Convergence curves of four algorithms in Medium-scale cases.

TABLE V
RESULTS OF HBS-VNS-SA ON ACTUAL CASES

Inst.	Total Delivery Time (h)	Maximum Delivery Time (h)	Number of Trucks	Obj (€)	CPU (s)
S-01	4.70	1.12	7	451.24	2.29
S-02	5.53	1.39	6	443.62	2.36
S-03	4.62	1.05	7	447.00	2.18
S-04	4.37	0.92	7	435.91	2.12
S-05	4.25	1.27	6	402.72	1.66
S-06	4.34	0.90	7	434.17	2.41
S-07	5.38	1.24	6	434.90	2.13
S-08	4.68	1.07	6	409.18	2.17
S-09	3.68	1.02	7	418.55	2.14
S-10	4.29	1.36	7	446.68	2.28
S-AVG	4.59	1.13	6.60	432.40	2.17
M-01	7.71	1.30	11	704.98	7.33
M-02	9.24	1.67	9	681.02	7.18
M-03	9.06	1.48	10	709.95	7.64
M-04	8.72	1.83	8	630.55	7.09
M-05	10.04	1.94	8	672.36	7.83
M-06	9.44	1.25	11	754.26	8.36
M-07	9.48	1.68	9	688.29	7.12
M-08	9.75	1.40	10	727.55	6.67
M-9	8.71	1.37	9	656.51	7.08
M-10	12.63	2.89	7	736.80	6.22
M-AVG	9.48	1.68	9.20	696.23	7.25
L-01	23.70	4.59	11	1272.18	26.60
L-02	21.73	4.68	8	1097.23	29.09
L-03	22.52	2.91	12	1227.08	23.81
L-04	18.29	2.01	14	1156.36	30.43
L-05	24.07	4.77	10	1248.44	32.53
L-06	19.49	2.47	13	1165.22	33.54
L-07	18.34	2.77	13	1140.59	32.37
L-08	20.49	3.26	13	1218.23	31.88
L-09	20.49	3.26	13	1218.23	28.47
L-10	21.69	2.99	13	1245.32	28.57
L-AVG	21.08	3.37	12.00	1198.89	29.73

In addition, to provide a comprehensive evaluation, we compare our proposed search algorithm with three classical search algorithms. The first algorithm is the Simulated Annealing (SA), where each operator only performs one random operation, such as swapping two random customers. With the initial temperature of 100 and the same cooling coefficient of 0.9, the termination temperature is set to 0.5, ensuring that the number of interactions is the same as that of our VNS-SA. The

second algorithm is the Variable Neighborhood Search (VNS). Unlike the VNS-SA, the VNS algorithm does not accept worse solutions during the search process. The maximum number of iterations of the VNS is set to 5000, and the algorithm would terminate if no better solution is obtained after exceeding 10 times. Finally, an adaptive large neighborhood search (ALNS) based on [34] is applied, including random destroy, cluster destroy, greedy repair and nearby-area repair.

Each algorithm was solved ten times for each case, the average objectives and average computation time are reported in Table VI. For consistency with the comparison in Section V-A.1, the values of parameters λ_1 , λ_2 , and λ_3 in objective function are set as 1.

It can be seen that the HBS-VNS-SA attains optimum performance in all cases. The HBS-VNS-SA can solve three different scale cases within 3s, 8s, and 30s, respectively, with a relatively stable computation time. In addition, it is evident that the computation time for HBS-VNS-SA-noSeg is notably longer compared to the HBS-VNS-SA. By calculating the feasible time of Segments in advance, HBS-VNS-SA can quickly judge the feasibility of new solutions obtained during the search. However, without Segments, the HBS-VNS-SA-noSeg needs to verify the feasibility of the entire route for all drone routes and customers during the solution validation. This significant difference in computation time proves the effectiveness of Segments.

Although the HBS-SA algorithms consume less time for computation, the solutions generated by HBS-SA are deemed unsatisfactory. This can be attributed to the use of relatively simple operators, thus the algorithm cannot explore the solution space thoroughly. Regarding the HBS-VNS algorithm, it sometimes finds solutions similar to HBS-VNS-SA (e.g., M-2, L-1), but it takes much longer computation time. Although the HBS-VNS terminates early in some cases, there still exists an apparent gap between the solutions found by the HBS-VNS and the HBS-VNS-SA. Additionally, HBS-VNS's computation time varies widely even for similar-sized problems, which implies a lack of robustness in the HBS-VNS. The computation time of the ALNS algorithm is comparable to that of HBS-VNS-SA, while its solution quality often falls short. A significant reason for this inferior performance lies

TABLE VI
COMPARISON RESULTS ON ACTUAL CASES

Inst.	HBS-VNS-SA		HBS-VNS-SA-noSeg		HBS-SA		HBS-VNS		HBS-ALNS	
	obj	CPU(s)	obj	CPU(s)	obj	CPU(s)	obj	CPU(s)	obj	CPU(s)
S-1	13.62	2.33	13.62	3.17	14.26	0.93	14.12	1.40	14.22	2.54
S-2	13.31	1.89	13.31	2.26	14.60	0.67	14.17	2.14	15.81	1.98
S-3	13.83	1.85	13.83	2.14	17.10	0.62	15.32	1.48	14.93	2.27
S-4	12.80	2.11	12.80	2.18	13.39	1.01	12.85	2.89	14.29	2.22
S-5	11.79	1.85	11.81	2.27	12.74	0.65	12.55	1.80	12.79	2.16
S-6	12.80	1.83	12.80	2.20	16.53	0.58	13.42	1.67	14.12	1.91
S-7	12.92	2.03	12.92	2.14	14.42	0.84	14.85	0.76	14.16	2.49
S-8	12.33	2.09	12.33	2.62	12.93	0.76	12.74	1.98	13.60	2.03
S-9	11.95	2.12	11.96	2.60	13.32	0.65	12.22	2.49	13.62	2.60
S-10	12.80	2.24	12.80	2.52	13.79	0.78	13.72	1.47	14.02	2.21
M-1	21.02	7.28	21.49	9.09	23.37	2.67	24.89	3.38	24.54	8.54
M-2	21.24	7.22	21.24	8.29	22.36	2.80	21.38	9.25	23.90	7.46
M-3	21.96	6.68	21.96	9.05	31.54	2.50	22.88	7.00	26.77	7.63
M-4	20.54	5.93	20.54	7.43	21.83	2.33	21.32	1.84	22.25	6.79
M-5	22.12	6.11	22.49	7.19	25.95	2.24	23.02	12.14	24.31	7.46
M-6	22.40	6.48	22.40	7.50	23.80	2.11	23.44	5.90	24.47	7.81
M-7	22.64	6.57	22.75	6.83	24.62	2.67	23.55	4.65	24.95	6.91
M-8	22.41	6.14	22.57	8.08	29.83	2.35	23.89	6.83	25.84	5.89
M-9	20.30	6.68	20.30	7.58	22.39	2.30	21.66	6.02	22.85	7.11
M-10	24.20	6.38	24.20	8.72	28.27	2.40	24.52	14.05	27.81	7.51
L-1	43.97	23.79	45.72	29.39	52.98	9.00	44.34	53.27	50.78	28.15
L-2	37.42	25.86	37.82	31.12	42.60	10.46	39.65	40.00	46.39	26.99
L-3	39.65	22.76	39.65	25.19	46.50	8.52	41.93	19.55	49.89	25.27
L-4	36.40	25.08	36.40	34.25	38.31	10.76	37.49	10.78	42.64	29.74
L-5	43.54	22.93	43.55	27.95	57.21	8.47	46.82	51.99	53.76	23.64
L-6	39.39	22.66	40.78	24.32	46.10	8.73	39.85	25.80	47.50	23.77
L-7	37.10	25.24	37.10	31.58	47.87	10.02	40.74	33.42	43.57	28.39
L-8	37.39	21.86	38.63	21.96	45.82	8.22	39.91	66.53	46.54	24.85
L-9	39.16	27.63	39.16	31.60	41.47	10.95	41.85	20.52	45.24	33.37
L-10	41.42	25.68	42.64	30.02	49.12	9.61	47.64	12.68	49.46	25.48

in the operators it employs, which are not tailored to the specific characteristics of the problem. Given the highly discrete solution space, it would be challenging for the ALNS to find optimal solutions efficiently. Through comparison, it can be proved that our designed operators can effectively explore the solution space and find better solutions. Moreover, our search framework exhibits excellent stability and effectiveness, enabling the discovery of superior solutions within an acceptable time.

Furthermore, the first six medium-scale cases are applied to generate convergence curves in Figure 11. Among these algorithms, the HBS-VNS exhibits the fastest convergence speed. However, it has a tendency to get stuck in local optima, which terminated the search prematurely in M-02 and M-04. Although the HBS-SA algorithm can consistently iterate and discover better solutions, its convergence rate is comparatively slow, and the solutions obtained within 5000 iterations are not of high quality. A similar issue occurs with the HBS-ALNS algorithm. Nevertheless, despite its slower convergence rate, the majority of cases do not converge within 5000 iterations due to the enhanced search framework of the ALNS. This suggests that continuous searching might lead to better outcomes. In contrast, the HBS-VNS-SA algorithm shows rapid convergence, which has the ability to escape local optima and find higher-quality solutions.

To explore the robustness of these algorithms, the “M-01” case is solved 100 times using four algorithms. The results

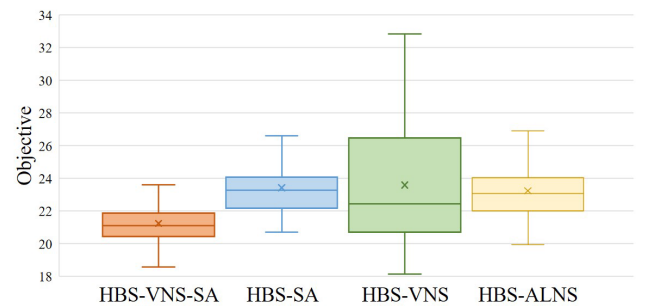


Fig. 12. Boxplots of four algorithms in a Medium-scale case (M-1).

are represented as boxplots in Fig. 12. It can be observed that among four algorithms, the HBS-VNS-SA has the best robustness, with its solutions more tightly clustered. Although the HBS-VNS can also produce some favorable solutions, it has the broadest range of solutions with higher median and mean values. This outcome further validates the effectiveness and robustness of the proposed HBS-VNS-SA algorithm.

C. Sensitivity Analysis

The sensitivity tests are conducted on a practical case, “M-05”, which has five critical factors: the average length of customer time windows, the service time of customers, the drone speed, the maximum capacity of the drone, and the capacity of the drone’s battery power.

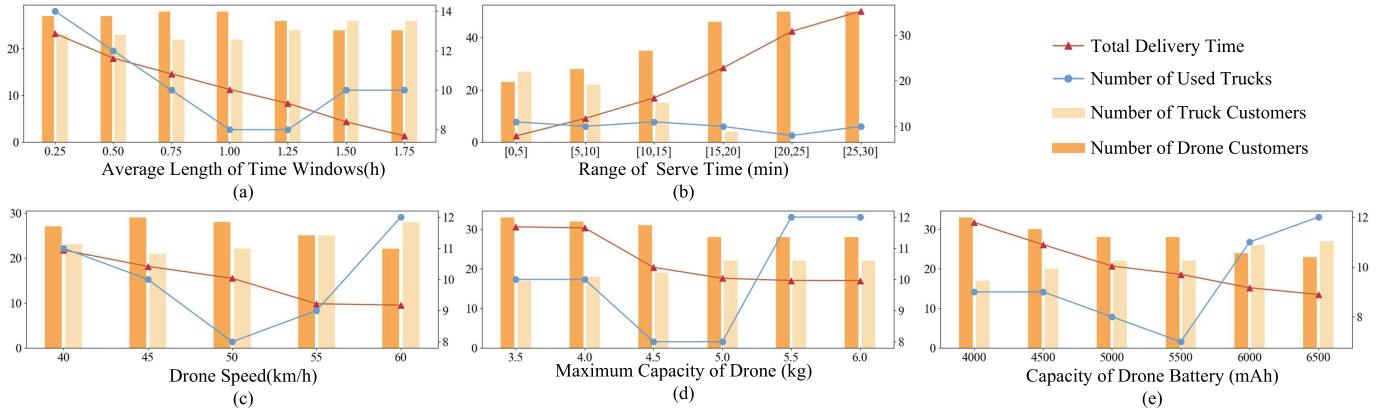


Fig. 13. Computational results under (a) Different lengths of time windows, (b) Different ranges of service time, (c) Different drone speeds, (d) Different maximum drone payloads, and (e) Different drone battery capacities.

1) *Impact Analysis for the Average Length of Customer Time Windows:* When generating the time windows for customers, an average length is defined to quantify the strictness of the time window. To monitor the effect of this parameter, comparison experiments are conducted with different values from 0.25h to 1.95h. The results are shown in Fig. 13(a).

As the time windows become looser, there is a significant reduction in the total delivery time, decreasing from 12.87h to 7.69h. When the time window length increases from 0.25h to 1.25h, the number of trucks used decreases from 14 to 8. However, more trucks are employed as the time window length increases beyond 1.25h. The reason may be that more relaxed time windows can provide more opportunities for cooperation delivery, and more drones are used to serve more customers. As for the number of customers served by trucks and drones, there is an overall trend of increasing drone customers as the relaxation degree of the time windows increases.

2) *Impact Analysis for the Service Time for Customers:* To investigate the impact of service time on customers, sensitivity tests are performed on the range of service time duration. Multiple intervals are considered, varying from [0min, 5min] to [25min, 30min]. The results are presented in Fig. 13(b). As the length of service time increases, the required delivery time increases significantly. When the customer service time exceeds 15 minutes, the overall delivery time would exceed 20 hours. Moreover, if the service time is more than 20 minutes, drones cannot provide service for any customers due to endurance limitations. However, the length of service time does not exert a significant effect on the number of trucks.

3) *Impact Analysis for the Speed of the Drone:* Unlike trucks constrained by ground traffic, drones have greater flexibility in their flight speed. In order to examine the impact of drone speed, a sensitivity analysis is conducted, varying the flight speed within the range of 40km/h to 60km/h.

The results in Fig. 13(c) reveal that, as the drone speed increases, the number of customers served by drones also increases, leading to a decrease in the overall delivery time. Notably, when the drone speed increases from 50 km/h to 55km/h, the most significant reduction in delivery time is observed, from 10.03h to 9.22h. However, with flight speed further improved, the reduction in overall delivery time becomes minimal, decreasing only by 0.04h. This suggests

that excessively high drone speeds are not advantageous for cooperation between two vehicles. For the current case, with trucks traveling at 40km/h, setting the drone speed within the 50-55km/h range is more appropriate.

4) *Impact Analysis for the Maximum Capacity of the Drone:* Improving the drone's payload capacity can make it visit more customers in a single flight, promoting the efficiency of drones. Therefore, the drone payload capacity is modified from 3.5kg to 6kg for the experiment, and the corresponding results are shown in Fig. 13(d). As the drone payload capacity increases, there is a decrease in the total delivery time. When the drone capacity increases to 4.5kg, eight vehicles are sufficient to complete all the deliveries, and the overall delivery time is greatly reduced to 10.38h. However, once the maximum payload capacity of the drone exceeds 5.5kg, additional increments in the maximum load cannot reduce delivery time. This is because the drone's endurance limits its ability to transport more parcels in a single flight.

5) *Impact Analysis for the Endurance of the Drone:* Enhancing the endurance of drones is also an approach to improve their efficiency. To assess the impact of drone endurance, the capacity of the drone battery is varied from 4000 mAh to 6500 mAh, and the results are shown in Figure 13(e).

As the battery capacity increases from 4000 mAh to 6500 mAh, a significant reduction in the total delivery time is observed, ranging from 11.29h to 8.89h. It can be inferred that increasing the battery power is a valid way to reduce the delivery time. Moreover, as the battery capacity increases from 4000 mAh to 5500 mAh, there is a consistent decrease in the number of trucks utilized. With improved endurance, drones are able to serve more customers even with limited nodes for take-off and landing. However, when drone endurance continues to increase, the number of trucks rapidly escalates to 11, in which assigning distant customers to drones becomes a priority to further reduce the overall delivery time.

6) *Management Insights:* The sensitivity analysis provides valuable insights for managers and policymakers in optimizing cooperated delivery systems with trucks and drones.

Firstly, relaxing customer time windows significantly reduces total delivery time, highlighting the importance of flexibility in scheduling. However, it is important to note

that excessively loose time windows may lead to decreased customer satisfaction and potentially increased delivery costs. Secondly, shortening service time is an effective means of reducing overall delivery time and enhancing service quality.

Regarding drone operational parameters, drone speed should be set in conjunction with truck speed. In this study where trucks travel at 40km/h, maintaining a drone speed of 55km/h is found to be the most reasonable for efficient cooperation. Furthermore, compared to increasing drone payload capacity, extending drone endurance has a greater impact on reducing overall delivery time. However, managers should also consider the trade-off between drone endurance and truck usage, as excessively high endurance may lead to an increase in both the number of trucks required and the overall delivery cost.

Overall, these insights underscore the practical significance of considering various time and drone-related factors in optimizing cooperative delivery systems.

VI. CONCLUSION

Recognizing the effectiveness and cost savings of cooperation between trucks and drones, the delivery routing problem of multiple trucks, each equipped with one drone, is studied. The impact of varying drone payloads on energy consumption and the customer time windows are considered, which puts higher requirements on the cooperation of two vehicles. The concept of Segment is introduced, and a heuristic based on Segment is designed, where the drone route and the corresponding truck route segment are constructed synchronously. The variable neighborhood search combined with simulated annealing is developed to improve the quality of the solutions.

The small-scale cases based on the Solomon benchmark are solved by both CPLEX and the proposed algorithm. The algorithm is also compared with another exact algorithm, demonstrating that it could find near-optimal solutions in less time. Furthermore, three scales of actual cases based on the road network of Changsha are constructed and solved by our algorithm, which further proves the effectiveness and robustness of the algorithm. Finally, sensitivity analysis is conducted on customer time preferences and drone performance parameters. It is proved that the looser time windows and the shorter service time would lead to a shorter overall delivery time. Besides, compared with increasing the drone payload, improving its endurance can help reduce more delivery time.

Recently, cooperated trucks and drones have been a hot issue. The problem in this paper can be further analyzed with more detailed costs, such as drone charging costs. The communication distances between drones and trucks during simultaneous movement could also be considered. In addition, uncertainty or dynamism can also be further explored, such as dynamic demands of customers. More exciting and valuable variants of this problem are waiting to be explored.

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